



LUNDS
UNIVERSITET

Matematikcentrum

Matematik NF

Tentamensskrivning
Topology
Wednesday May 29, 2013
Time: 08.00–13.00

No aids allowed. Use the distributed paper sheets and write only on one side, with at most one problem per page. Fill in the cover sheet completely and initialize every sheet. Write legibly (in Swedish or English). Motivate your conclusions clearly and concisely; draw a picture if that helps to clarify your argument.

Test results: Will be posted on the course home page before 17.00 Saturday June 1. For oral exams, contact the course leader.

1. Let

$$\mathcal{E} = \{ p(x^4, y^5) : p(r, s) \text{ real polynomial in } (r, s) \in \mathbf{R}^2 \}$$

- Show that $\mathcal{E}$ is dense in $\mathcal{C}(X, \mathbf{R})$, where $X = [0, 1] \times [-1, 1]$, with the usual topology.
 - Is $\mathcal{E}$ dense in $\mathcal{C}(Y, \mathbf{R})$ when $Y = [-1, 1] \times [-1, 1]$? Motivate your answer.
2. Let $\mathcal{T}$ be a family of sets $U \subseteq \mathbf{R}$ with the property that the complement $\mathbf{R} \setminus U$ is countable, and we also include the empty set in $\mathcal{T}$.
- Show that $\mathcal{T}$ gives a topology on $\mathbf{R}$.
 - Show that $(\mathbf{R}, \mathcal{T})$ is T_1 but not a Hausdorff space.
 - What is the closure of $(-1, 1)$ in the topology given by $\mathcal{T}$?
 - Show that $(\mathbf{R}, \mathcal{T})$ is not separable.

3. Let X be the set of all subsets of the positive integers $\mathbf{N}$. For $A, B \in X$ define

$$d(A, B) = \sum_{n \in A \triangle B} \frac{1}{n^3}$$

(defined as 0 when $A = B$) where $A \triangle B = (A \cup B) \setminus (A \cap B)$ is the symmetric difference of A and B , i.e., the set of points n belonging to exactly one of A and B . Show that d is a metric on X and that the metric space (X, d) is separable.

4. Assume that $f : X \rightarrow Y$ is continuous, where X is compact and Y is Hausdorff. Show that if $A_1 \supseteq A_2 \supseteq A_3 \supseteq \dots$ are closed subsets of X then

$$f\left(\bigcap_{n=1}^{\infty} A_n\right) = \bigcap_{n=1}^{\infty} f(A_n)$$

5. A topological space X is said to be countable compact if every countable open cover of X has a finite subcover. Show that a countable compact metric space X is compact. (Observe that we do not assume that X is second countable.)